

FUNDACIÓN FULGOR

Undergrad CMOS Course

Homework N°5

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1 Introduction

In this report, we will present the solutions to the analog *CMOS* design exercises proposed in *Homework 5* for the Undergrad *CMOS* Course. Unless specific parameter values are provided within the problem statements, we will adhere to the general assumptions given in the assignment:

$$\begin{cases} \mu_n \cdot C_{ox} = 200 \ \left[\frac{\mu A}{V^2}\right] \\ \mu_p \cdot C_{ox} = 100 \ \left[\frac{\mu A}{V^2}\right] \\ V_{Th,n} = 0.4\,[V] \\ V_{Th,p} = -0.4\,[V] \\ \lambda = 0 \end{cases} \tag{1}$$

2 Exercise 1: Common-Source Bias Design

2.1 Problem Statement

The objective of this exercise is to design the common-source circuit shown in Fig. 1 for a drain current of $I_D = 1\,[mA]$.

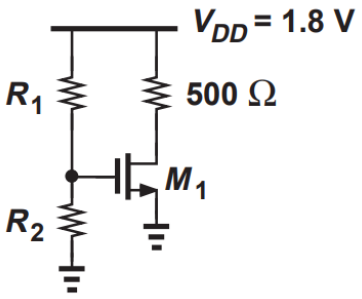


Figure 1: Exercise 1 (Problem 7.2). [1, p. 340]

Given the transistor aspect ratio $W/L = 20/0.18$, we must calculate the bias resistors R_1 and R_2 such that the input impedance is at least $20\,[k\Omega]$. The technology parameters are $V_{DD} = 1.8\,[V]$ and the rest must be followed according to Eq. 1

2.2 Analysis and Resolution

First, we assume the transistor M_1 operates in the saturation region. Since channel-length modulation is neglected ($\lambda = 0$), the drain current is given by:

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{Th,n})^2 \tag{2}$$

Substituting the given values into the equation:

$$1 \times 10^{-3} = \frac{1}{2}(200 \times 10^{-6}) \left(\frac{20}{0.18}\right) (V_{GS} - 0.4)^2 \tag{3}$$

$$1 \times 10^{-3} \approx 11.11 \times 10^{-3}(V_{GS} - 0.4)^2 \implies (V_{GS} - 0.4)^2 \approx 0.09 \tag{4}$$

Solving for V_{GS}, we obtain $V_{GS} = 0.7\,[V]$. Since the source is grounded, the gate voltage must be $V_G = 0.7\,[V]$. Notice that the gate voltage is established by the R_1-R_2 voltage divider, this is:

$$V_G = V_{DD}\frac{R_2}{R_1 + R_2} \implies 0.7 = 1.8\frac{R_2}{R_1 + R_2} \tag{5}$$

This yields the relationship shown in Eq. 6.

$$R_1 \approx 1.57R_2 \tag{6}$$

To satisfy the input impedance requirement ($R_{in} \geq 20\,[k\Omega]$), we evaluate the parallel combination of the bias resistors:

$$R_{in} = R_1//R_2 = \frac{1.57R_2 \cdot R_2}{1.57R_2 + R_2} \approx 0.611R_2 \tag{7}$$

$$0.611R_2 \geq 20\,[k\Omega] \implies R_2 \geq 32.73\,[k\Omega] \tag{8}$$

Selecting the minimum allowable values to satisfy the design constraint, we obtain:

$$\begin{cases} R_2 = 32.73\,[k\Omega] \\ R_1 = 51.43\,[k\Omega] \end{cases} \tag{9}$$

Note: To verify saturation, we check the drain voltage: $V_D = V_{DD} - I_D R_D = 1.8 - (1\,[mA])(500\,[\Omega]) = 1.3\,[V]$. *Since* $V_D > V_G - V_{TH,n}$ *(*$1.3\,[V] > 0.3\,[V]$*), the assumption holds true.*

3 Exercise 2: Common-Source with Source Degeneration

3.1 Problem Statement

The circuit shown in Fig. 2 must be designed for a voltage drop of $V_{RS} = 200\,[mV]$ across R_S.

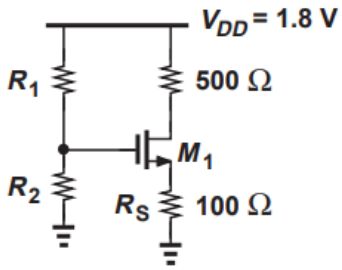


Figure 2: Exercise 2 (Problem 7.4). [1, p. 341]

The specific requirements are:

(a) Calculate the minimum allowable value of W/L if M_1 must remain in saturation.

(b) Calculate the required values of R_1 and R_2 if the input impedance must be at least $30\,[k\Omega]$.

The technology parameters are given by Eq. 1, along with $V_{DD} = 1.8\,[V]$, $R_D = 500\,[\Omega]$, and $R_S = 100\,[\Omega]$ as shown in Fig. 2.

3.2 Analysis and Resolution

3.2.1 Part (a): Minimum allowable value of W/L

First, we determine the drain current I_D flowing through the circuit, which can be calculated using the specified voltage drop across the source resistor R_S:

$$I_D = \frac{V_{RS}}{R_S} = \frac{0.2\,[V]}{100\,[\Omega]} = 2\,[mA] \tag{10}$$

To find the saturation condition, we need the drain voltage V_D and the source voltage V_S. The source voltage is directly given as $V_S = V_{RS} = 0.2\,[V]$. The drain voltage is calculated as follows:

$$V_D = V_{DD} - I_D R_D = 1.8 - (2 \times 10^{-3})(500) = 0.8\,[V] \tag{11}$$

Consequently, the drain-to-source voltage V_{DS} is:

$$V_{DS} = V_D - V_S = 0.8 - 0.2 = 0.6\,[V] \tag{12}$$

For the transistor to remain in saturation, the condition $V_{DS} \geq V_{GS} - V_{Th,n}$ must be met. Thus, the maximum allowable overdrive voltage (The term $(V_{GS} - V_{Th,n})$ is defined as the overdrive voltage V_{OV}) for the transistor to stay in saturation is:

$$V_{OV,max} = V_{DS} = 0.6\,[V] \tag{13}$$

Using the drain current equation for saturation (Eq. 2), we solve for the minimum aspect ratio $(W/L)_{min}$ using the maximum overdrive voltage:

$$I_D = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_{min} V_{OV,max}^2 \tag{14}$$

$$2 \times 10^{-3} = \frac{1}{2}(200 \times 10^{-6}) \left(\frac{W}{L}\right)_{min} (0.6)^2 \tag{15}$$

$$2 \times 10^{-3} = 36 \times 10^{-6} \left(\frac{W}{L}\right)_{min} \implies \left(\frac{W}{L}\right)_{min} = \frac{2000}{36} \approx 55.56 \tag{16}$$

3.2.2 Part (b): Required values of R_1 and R_2

Using the maximum overdrive voltage established in subsubsection 3.2.1, we calculate the required gate-to-source voltage V_{GS}:

$$V_{GS} = V_{OV,max} + V_{Th,n} = 0.6 + 0.4 = 1.0\,[V] \tag{17}$$

Since $V_S = 0.2\,[V]$, the absolute gate voltage V_G referenced to ground must be:

$$V_G = V_{GS} + V_S = 1.0 + 0.2 = 1.2\,[V] \tag{18}$$

The gate voltage is set by the resistive divider formed by R_1 and R_2:

$$V_G = V_{DD}\frac{R_2}{R_1 + R_2} \implies 1.2 = 1.8\frac{R_2}{R_1 + R_2} \tag{19}$$

$$1.2(R_1 + R_2) = 1.8R_2 \implies 1.2R_1 = 0.6R_2 \implies R_2 = 2R_1 \tag{20}$$

The input impedance requirement is $R_{in} \geq 30\,[k\Omega]$. Evaluating the parallel combination:

$$R_{in} = R_1 // R_2 = \frac{R_1(2R_1)}{R_1 + 2R_1} = \frac{2}{3}R_1 \tag{21}$$

$$\frac{2}{3}R_1 \geq 30\,[k\Omega] \implies R_1 \geq 45\,[k\Omega] \tag{22}$$

Using the lower bound to satisfy the constraint, and applying the relationship from Eq. 20, we determine the required resistor values:

$$\begin{cases} R_1 = 45\,[k\Omega] \\ R_2 = 90\,[k\Omega] \end{cases} \tag{23}$$

4 Exercise 3: Self-Biased Stage Design

4.1 Problem Statement

The self-biased stage shown in Fig. 3 must be designed for a drain current of $I_D = 1\,[mA]$.

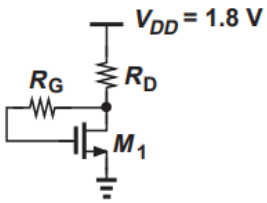


Figure 3: Exercise 3 (Problem 7.6). [1, p. 341]

If transistor M_1 must provide a transconductance of $g_m = 1/100\,[\Omega^{-1}]$), we need to calculate the required value of R_D. The gate resistor R_G must be neglected for DC analysis. The general technology parameters from Eq. 1 apply, along with $V_{DD} = 1.8\,[V]$.

4.2 Analysis and Resolution

First, we determine the required transconductance g_m in standard units:

$$g_m = \frac{1}{100\,[\Omega]} = 0.01\left[\frac{A}{V}\right] = 10\left[\frac{mA}{V}\right] \tag{24}$$

In the saturation region, the transconductance can be expressed in terms of the drain current I_D and the overdrive voltage $(V_{GS} - V_{Th,n})$:

$$g_m = \frac{2I_D}{V_{GS} - V_{Th,n}} \tag{25}$$

By rearranging Eq. 25, we can solve for the required gate-to-source voltage V_{GS}:

$$V_{GS} - V_{Th,n} = \frac{2I_D}{g_m} = \frac{2 \times 10^{-3}\,[A]}{10 \times 10^{-3}\,[A/V]} = 0.2\,[V] \tag{26}$$

$$V_{GS} = 0.2\,[V] + V_{Th,n} = 0.2\,[V] + 0.4\,[V] = 0.6\,[V] \tag{27}$$

For the DC bias analysis of this self-biased configuration, we must consider the voltage drop across the feedback resistor R_G. Since the DC gate current of a $MOSFET$ is ideally zero ($I_G = 0$), there is no voltage drop across R_G. Therefore, the drain voltage V_D is equal to the gate voltage V_G. Since the source is grounded ($V_S = 0$), we conclude that:

$$V_D = V_G = V_{GS} = 0.6\,[V] \tag{28}$$

Finally, the required drain resistor R_D can be calculated using *Ohm's law*, given the voltage drop across it and the drain current flowing through it:

$$R_D = \frac{V_{DD} - V_D}{I_D} \tag{29}$$

$$R_D = \frac{1.8\,[V] - 0.6\,[V]}{1 \times 10^{-3}\,[A]} = \frac{1.2\,[V]}{1 \times 10^{-3}\,[A]} = 1.2\,[k\Omega] \tag{30}$$

Thus, the required value for the drain resistor is $R_D = 1.2\,[k\Omega]$. Note that the condition $V_{DS} = V_{GS}$ inherently guarantees that the transistor operates in the saturation region, as $V_{DS} > V_{GS} - V_{Th,n}$ is strictly satisfied ($0.6\,[V] > 0.2\,[V]$).

5 Exercise 4: Common-Source Stage Analysis

5.1 Problem Statement

In the common-source stage shown in Fig. 4, the transistor has an aspect ratio of $W/L = 30/0.18$ and channel-length modulation is neglected ($\lambda = 0$).

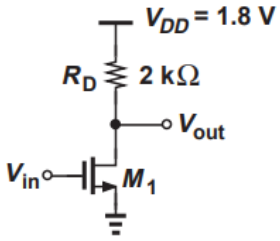


Figure 4: Exercise 4 (Problem 7.17). [1, p. 342]

We are tasked with determining:

(a) The gate voltage V_G that yields a drain current of $I_D = 0.5\,[mA]$, and verifying that M_1 operates in saturation.

(b) The voltage gain A_v of the stage under this bias condition.

The parameters given are $V_{DD} = 1.8\,[V]$ and $R_D = 2\,[k\Omega]$. We will use the general technology parameters from Eq. 1.

5.2 Analysis and Resolution

5.2.1 Part (a): Gate Voltage and Saturation Verification

Assuming the transistor operates in the saturation region, the drain current is described by Eq. 2. We can rearrange this equation to solve for the required overdrive voltage $V_{OV} = (V_{GS} - V_{Th,n})$:

$$V_{OV} = \sqrt{\frac{2I_D}{\mu_n C_{ox}\left(\frac{W}{L}\right)}} \tag{31}$$

Substituting the known values:

$$V_{OV} = \sqrt{\frac{2(0.5 \times 10^{-3})}{(200 \times 10^{-6})\left(\frac{30}{0.18}\right)}} = \sqrt{\frac{10^{-3}}{(200 \times 10^{-6})(166.67)}} = \sqrt{\frac{10^{-3}}{33.33 \times 10^{-3}}} \tag{32}$$

$$V_{OV} = \sqrt{0.03} \approx 0.173\,[V] \tag{33}$$

Since the source is grounded ($V_S = 0$), the gate-to-source voltage V_{GS} is equal to the gate voltage V_G. Thus:

$$V_G = V_{GS} = V_{OV} + V_{Th,n} = 0.173\,[V] + 0.4\,[V] = 0.573\,[V] \tag{34}$$

To verify that M_1 operates in saturation, we must ensure that $V_{DS} \geq V_{OV}$. The drain voltage V_D is calculated by analyzing the voltage drop across R_D:

$$V_D = V_{DD} - I_D R_D = 1.8\,[V] - (0.5\times10^{-3}\,[A])(2\times10^{3}\,[\Omega]) = 1.8\,[V] - 1.0\,[V] = 0.8\,[V] \tag{35}$$

Since $V_S = 0$, $V_{DS} = 0.8\,[V]$. Comparing V_{DS} and V_{OV}:

$$0.8\,[V] > 0.173\,[V] \implies V_{DS} > V_{OV} \tag{36}$$

The condition is satisfied; therefore, M_1 operates in the saturation region.

5.2.2 Part (b): Voltage Gain

The small-signal voltage gain A_v of a common-source stage with a resistive load and $\lambda = 0$ (implying output resistance $r_o \to \infty$) is given by:

$$A_v = -g_m R_D \tag{37}$$

First, we determine the transconductance g_m of the transistor at the current bias point:

$$g_m = \frac{2I_D}{V_{OV}} = \frac{2(0.5 \times 10^{-3}\,[A])}{0.173\,[V]} = \frac{1 \times 10^{-3}}{0.173} \approx 5.77\ \left[\frac{mA}{V}\right] \tag{38}$$

Now, substituting g_m and R_D into the gain equation:

$$A_v = -(5.77 \times 10^{-3}\,[A/V])(2000\,[\Omega]) \approx -11.54\ \left[\frac{V}{V}\right] \tag{39}$$

6 Exercise 5: Common-Source Stage with Active Load

6.1 Problem Statement

The common-source stage shown in Fig. 5 must provide a voltage gain of $|A_v| = 10$ with a bias current of $I_D = 0.5\,[mA]$.

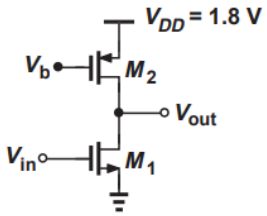


Figure 5: Exercise 5 (Problem 7.20). [1, p. 343]

For this problem, channel-length modulation must be considered, with parameters $\lambda_1 = 0.1\,[V^{-1}]$ and $\lambda_2 = 0.15\,[V^{-1}]$. We are asked to:

(a) Calculate the required aspect ratio $(W/L)_1$ of the input transistor M_1.

(b) Determine the required bias voltage V_b if $(W/L)_2 = 20/0.18$.

General technology parameters apply defined in Eq. 1 and Fig. 5.

6.2 Analysis and Resolution

6.2.1 Part (a): Required $(W/L)_1$

The magnitude of the voltage gain for a common-source stage with an active *PMOS* load is given by:

$$|A_v| = g_{m1}(r_{o1}//r_{o2}) \tag{40}$$

where g_{m1} is the transconductance of M_1, and r_{o1} and r_{o2} are the output resistances of M_1 and M_2, respectively.

First, we calculate the output resistances r_{o1} and r_{o2} using the bias current I_D:

$$r_{o1} = \frac{1}{\lambda_1 I_D} = \frac{1}{(0.1\,[V^{-1}])(0.5 \times 10^{-3}\,[A])} = 20\,[k\Omega] \tag{41}$$

$$r_{o2} = \frac{1}{\lambda_2 I_D} = \frac{1}{(0.15\,[V^{-1}])(0.5 \times 10^{-3}\,[A])} \approx 13.33\,[k\Omega] \tag{42}$$

The equivalent output resistance of the stage R_{out} is the parallel combination of r_{o1} and r_{o2}:

$$R_{out} = r_{o1}//r_{o2} = \frac{20\,[k\Omega] \times 13.33\,[k\Omega]}{20\,[k\Omega] + 13.33\,[k\Omega]} = 8\,[k\Omega] \tag{43}$$

Given the required voltage gain $|A_v| = 10$, we can solve for the necessary transconductance g_{m1}:

$$|A_v| = g_{m1}R_{out} \implies g_{m1} = \frac{10}{8000\,[\Omega]} = 1.25\,\left[\frac{mA}{V}\right] \tag{44}$$

The transconductance of M_1 in saturation can also be expressed as:

$$g_{m1} = \sqrt{2\mu_n C_{ox}\left(\frac{W}{L}\right)_1 I_D} \tag{45}$$

Squaring both sides and solving for $(W/L)_1$:

$$g_{m1}^2 = 2\mu_n C_{ox}\left(\frac{W}{L}\right)_1 I_D \implies \left(\frac{W}{L}\right)_1 = \frac{g_{m1}^2}{2\mu_n C_{ox} I_D} \tag{46}$$

$$\left(\frac{W}{L}\right)_1 = \frac{(1.25\times10^{-3})^2}{2(200\times10^{-6})(0.5\times10^{-3})} = \frac{1.5625\times10^{-6}}{200\times10^{-9}} \approx 7.81 \tag{47}$$

6.2.2 Part (b): Required Bias Voltage V_b

To find the bias voltage V_b at the gate of the *PMOS* transistor M_2, we use its drain current equation. For *DC* biasing purposes, we can neglect the small variations caused by channel-length modulation:

$$I_D = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_2 |V_{GS2} - V_{Th,p}|^2 \tag{48}$$

Substituting the known parameters:

$$0.5\times10^{-3} = \frac{1}{2}(100\times10^{-6})\left(\frac{20}{0.18}\right)|V_{GS2} - (-0.4)|^2 \tag{49}$$

$$0.5\times10^{-3} = (50\times10^{-6})(111.11)|V_{GS2}+0.4|^2 = 5.555\times10^{-3}|V_{GS2}+0.4|^2 \tag{50}$$

$$|V_{GS2}+0.4|^2 = \frac{0.5\times10^{-3}}{5.555\times10^{-3}} \approx 0.09\,[V^2] \tag{51}$$

Taking the square root, we obtain the absolute value of the overdrive voltage for the *PMOS*:

$$|V_{GS2}+0.4| = \sqrt{0.09} = 0.3\,[V] \tag{52}$$

For a *PMOS* device, the source-to-gate voltage V_{SG2} must overcome the threshold voltage magnitude $|V_{Th,p}|$ to maintain operation:

$$V_{SG2} = |V_{GS2}+0.4| + |V_{Th,p}| = 0.3\,[V] + 0.4\,[V] = 0.7\,[V] \tag{53}$$

Since the source of M_2 is connected to V_{DD}, the bias voltage V_b (which is the gate

voltage V_{G2}) is calculated as:

$$V_{SG2} = V_{DD} - V_b \implies 0.7\,[V] = 1.8\,[V] - V_b \tag{54}$$

$$V_b = 1.8\,[V] - 0.7\,[V] = 1.1\,[V] \tag{55}$$

Thus, the required bias voltage is $V_b = 1.1\,[V]$.

7 Exercise 6: Common-Source Stage with Diode-Connected Load

7.1 Problem Statement

The objective of this final exercise is to design the circuit shown in Fig. 6 for a voltage gain of $|A_v| = 3$.

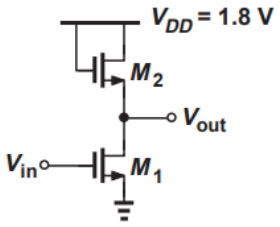


Figure 6: Exercise 6 (Problem 7.25). [1, p. 344]

Given the aspect ratio of the input transistor $(W/L)_1 = 20/0.18$, we must determine the required aspect ratio of the diode-connected load $(W/L)_2$. We assume channel-length modulation is negligible ($\lambda = 0$). The standard technology parameters from Eq. 1 apply.

7.2 Analysis and Resolution

For a common-source amplifier with a diode-connected *PMOS* load, and neglecting channel-length modulation ($\lambda = 0 \implies r_o \to \infty$), the small-signal voltage gain A_v is given by the ratio of the transconductances:

$$A_v = -\frac{g_{m1}}{g_{m2}} \implies |A_v| = \frac{g_{m1}}{g_{m2}} \tag{56}$$

The transconductance of a *MOSFET* in saturation can be expressed in terms of the bias current I_D and its physical parameters:

$$g_m = \sqrt{2\mu C_{ox}\left(\frac{W}{L}\right) I_D} \tag{57}$$

Substituting the transconductance expressions for both M_1 (*NMOS*) and M_2 (*PMOS*) into the gain equation:

$$|A_v| = \frac{\sqrt{2\mu_n C_{ox}(W/L)_1 I_D}}{\sqrt{2\mu_p C_{ox}(W/L)_2 I_D}} \tag{58}$$

Since both transistors are connected in series, they share the exact same *DC* drain current ($I_{D1} = I_{D2} = I_D$). We can cancel $\sqrt{2}$ and $\sqrt{I_D}$ from the numerator and denomi-

nator:

$$|A_v| = \sqrt{\frac{\mu_n C_{ox}(W/L)_1}{\mu_p C_{ox}(W/L)_2}} \tag{59}$$

Now, we square both sides to isolate and solve for the unknown aspect ratio $(W/L)_2$:

$$|A_v|^2 = \frac{\mu_n C_{ox}(W/L)_1}{\mu_p C_{ox}(W/L)_2} \tag{60}$$

$$\left(\frac{W}{L}\right)_2 = \frac{\mu_n C_{ox}(W/L)_1}{\mu_p C_{ox}|A_v|^2} \tag{61}$$

Substituting the known values ($|A_v| = 3$, $\mu_n C_{ox} = 200\,[\mu A/V^2]$, $\mu_p C_{ox} = 100\,[\mu A/V^2]$, and $(W/L)_1 = 20/0.18$):

$$\left(\frac{W}{L}\right)_2 = \frac{200 \times 10^{-6}\left(\frac{20}{0.18}\right)}{100 \times 10^{-6}(3)^2} \tag{62}$$

$$\left(\frac{W}{L}\right)_2 = \frac{2\left(\frac{20}{0.18}\right)}{9} = \frac{40}{0.18 \times 9} = \frac{40}{1.62} \approx 24.69 \tag{63}$$

Therefore, the required aspect ratio for the *PMOS* diode-connected load is approximately 24.69.

8 Conclusion

In conclusion, this report successfully presented the step-by-step design and analysis of fundamental analog *CMOS* amplifier stages, primarily focusing on various configurations of the common-source topology. Through the resolution of these six exercises, theoretical concepts were applied to determine appropriate *DC* bias conditions, verify transistor operation in the saturation region, and calculate key small-signal performance metrics such as voltage gain and input impedance.

The progression from simple resistive loads and source degeneration to more complex active and diode-connected *PMOS* loads provided valuable insight into the practical trade-offs of analog *IC* design. Furthermore, the systematic application of the provided technology parameters ensured a rigorous and consistent mathematical evaluation across all the proposed circuits, reinforcing the analytical foundation required for advanced *CMOS* design.

9 Bibliography

[1] B. Razavi, *Fundamentals of Microelectronics*, 2nd Edition. John Wiley & Sons, 2013, ISBN: 978-1-118-15632-2.